

CREDIT RISK SCORECARD

Artha Finance — A Machine Learning Approach to NBFC Loan Approval Policy

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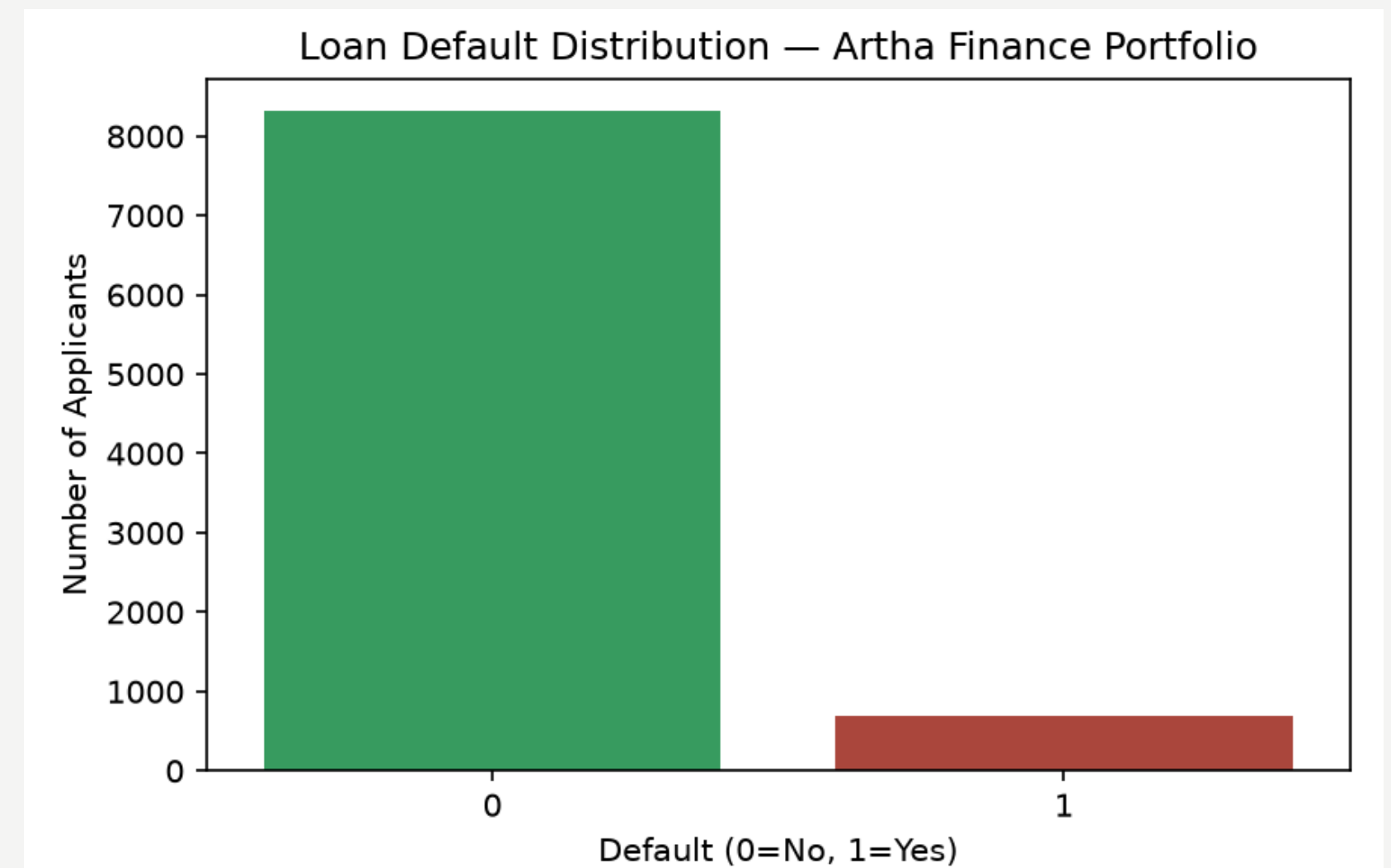
github.com/r33w33/credit-risk-scorecard-nbfc-

BUSINESS CONTEXT & DATA OVERVIEW

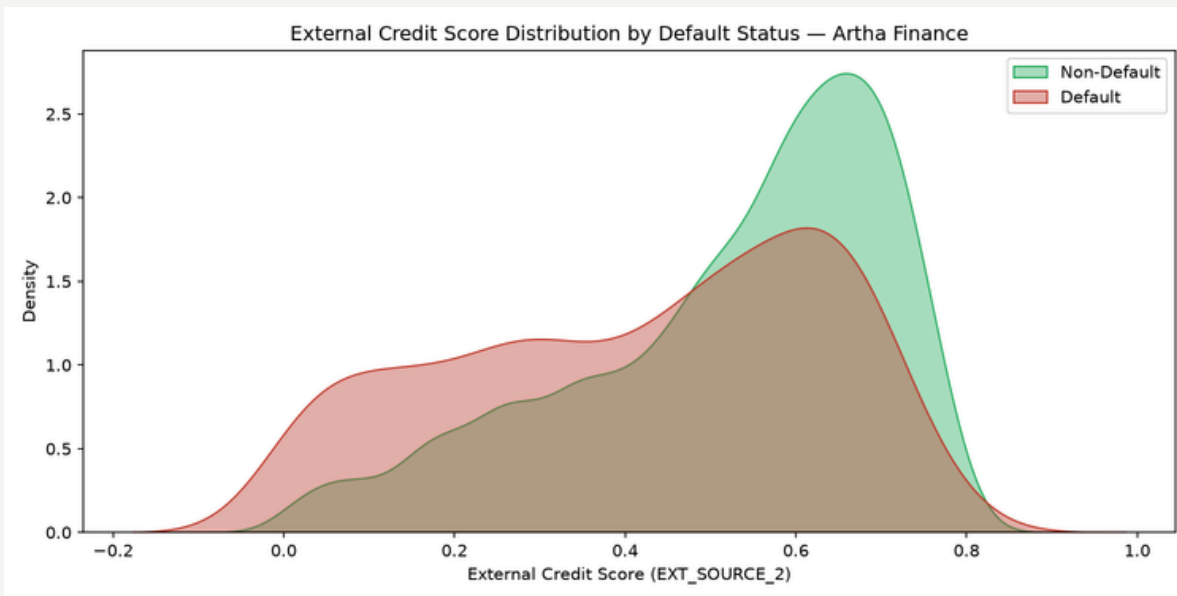
Artha Finance is a mid-size NBFC operating across Tier 2 cities in UP, Bihar, and MP. With a default rate of 8.3% against an industry average of 5.1%, the firm needed a data-driven approval framework to replace its current rule-based system that used only income and employment type as filters.

Dataset: 9,000 loan applications | 122 features | 7.63% default rate

Source: Home Credit Default Risk (Kaggle)

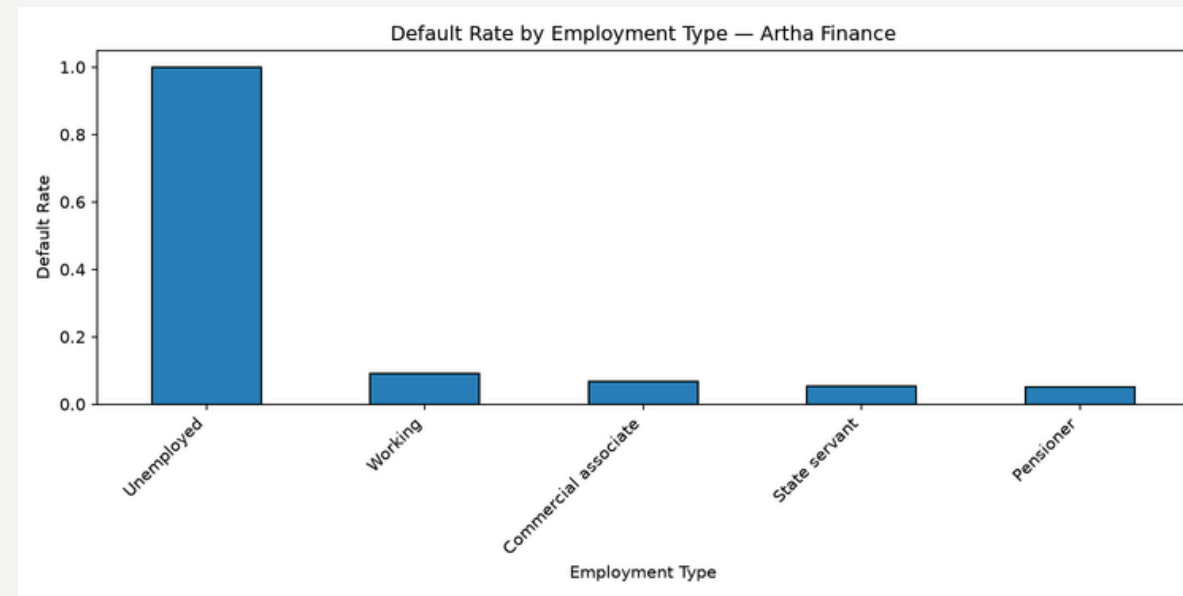


KEY EDA FINDINGS



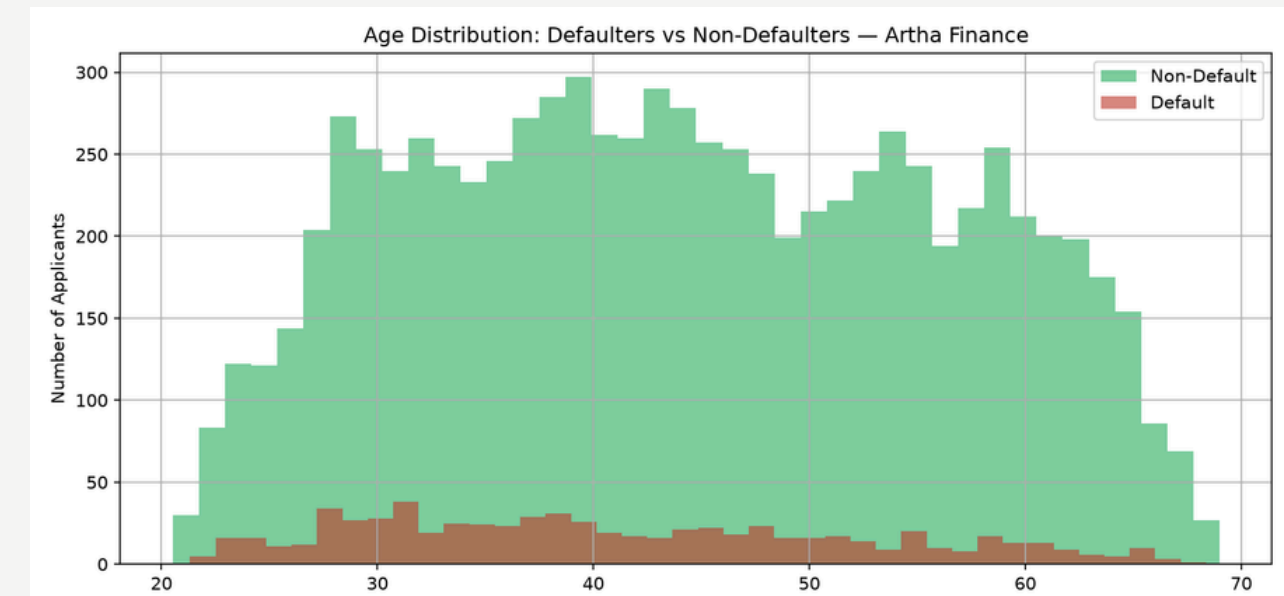
External Credit Score Distribution

External credit score is the strongest single predictor — defaulters cluster at lower scores



Default Rate by Employment Type

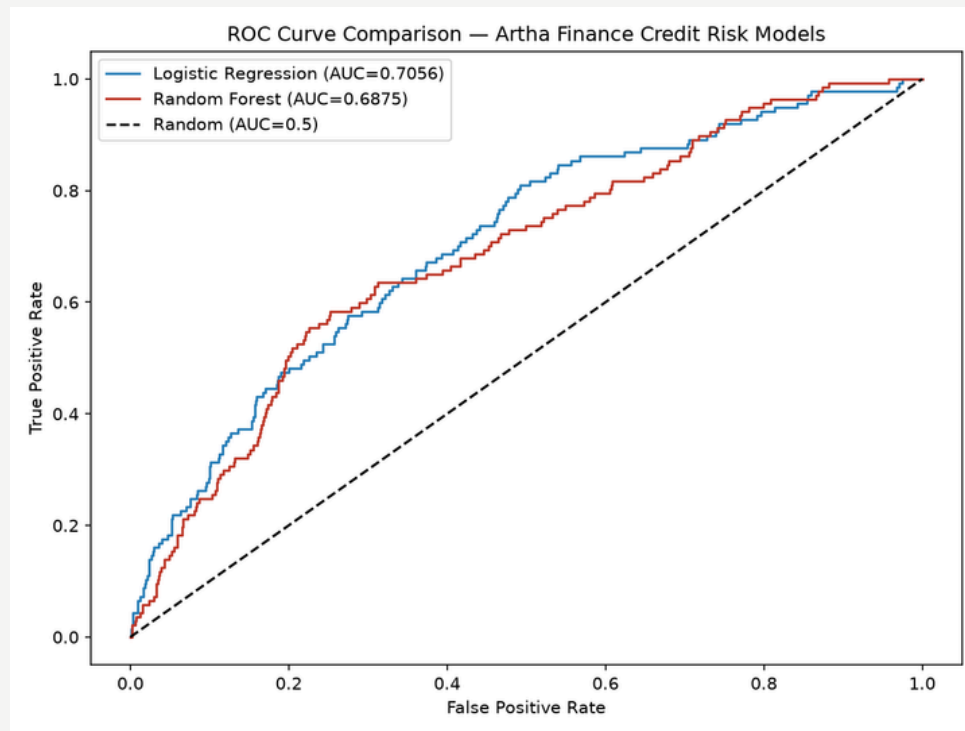
Unemployed applicants default at the highest rate — near-automatic decline signal



Age Distribution by Default Status

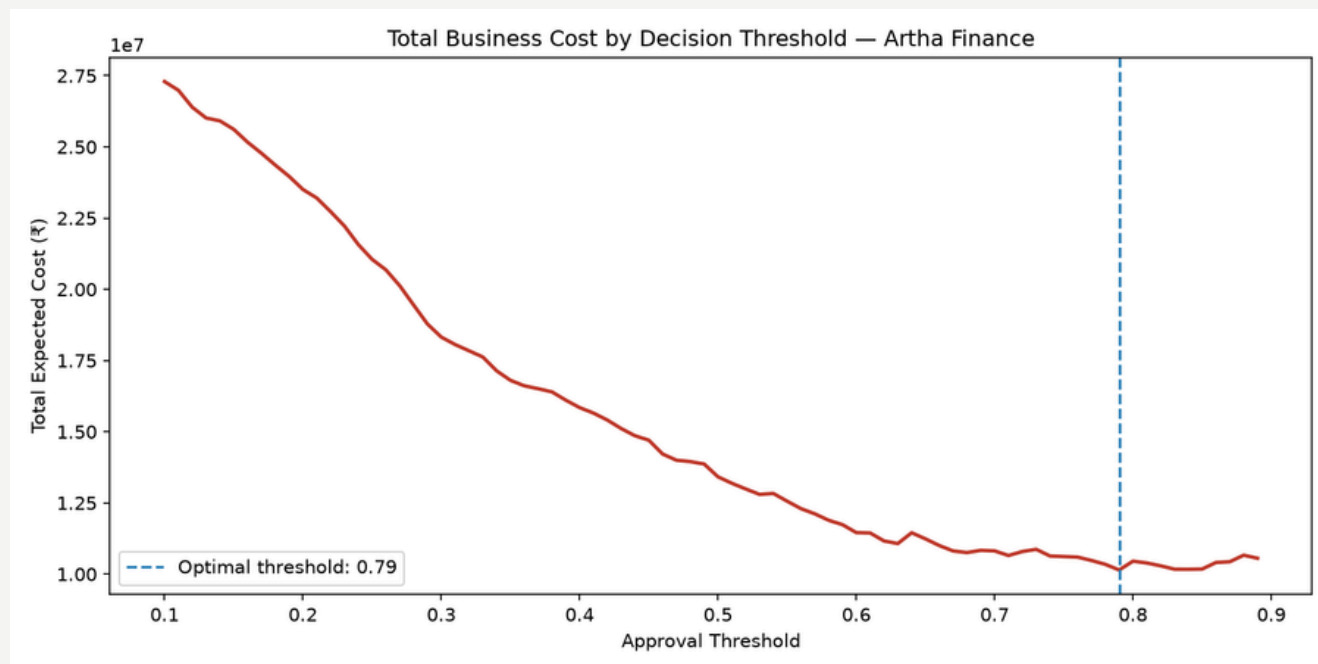
Younger applicants (20-30) show higher default concentration

MODEL RESULTS



Logistic Regression — AUC: 0.7056

Random Forest — AUC: 0.6875



Logistic Regression outperforms
Random Forest on this dataset.

Optimal approval threshold: 0.79
— minimises total expected
business cost at ₹1.01 crore on the
test set.

RISK SCORECARD & RECOMMENDATIONS

RISK BAND POLICY

GREEN (2.5% default) — Approve | Interest: 12-14% | No conditions

AMBER (3.2% default) — Approve with conditions | Interest: 15-17% | Salary slip + co-applicant

RED (8.5% default) — Manual review | Interest: 18-20% | Senior sign-off required

BLACK (17% default) — Decline | Reapply after 6 months

HIDDEN OPPORTUNITY

574 out of 593 AMBER-band applicants did not default (96.8%).

Artha Finance is leaving ₹15.99 crore in interest income unrealised by over-penalising this segment.

LIMITATIONS & NEXT STEPS

- Model trained on 9,000 applications — retrain on 50,000+ for production use
- Add SHAP explainability analysis for regulatory compliance
- Integrate bureau balance tables for richer credit history features